

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

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In [23]: # Playing with b
d=5
a=5

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=False)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (Look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+15)

# Draw different parts of the face and save in different frames

# Frame 1: draw the face:

make_closed_fig(plot, a=6, b=1.1, h=0, k=-5, d=d, num_of_points=20, color="white")
plot.plot_funs(plot_title="Playing with thehead", img_name="frame1.png")

make_closed_fig(plot, a=1, b=b, h=3.5, k=5, d=1, num_of_points=20, color='#FFC0CB')
make_closed_fig(plot, a=1, b=b, h=-3.5, k=5, d=1, num_of_points=20, color='pink')
plot.plot_funs(plot_title="Playing with ears", img_name="frame2.png")

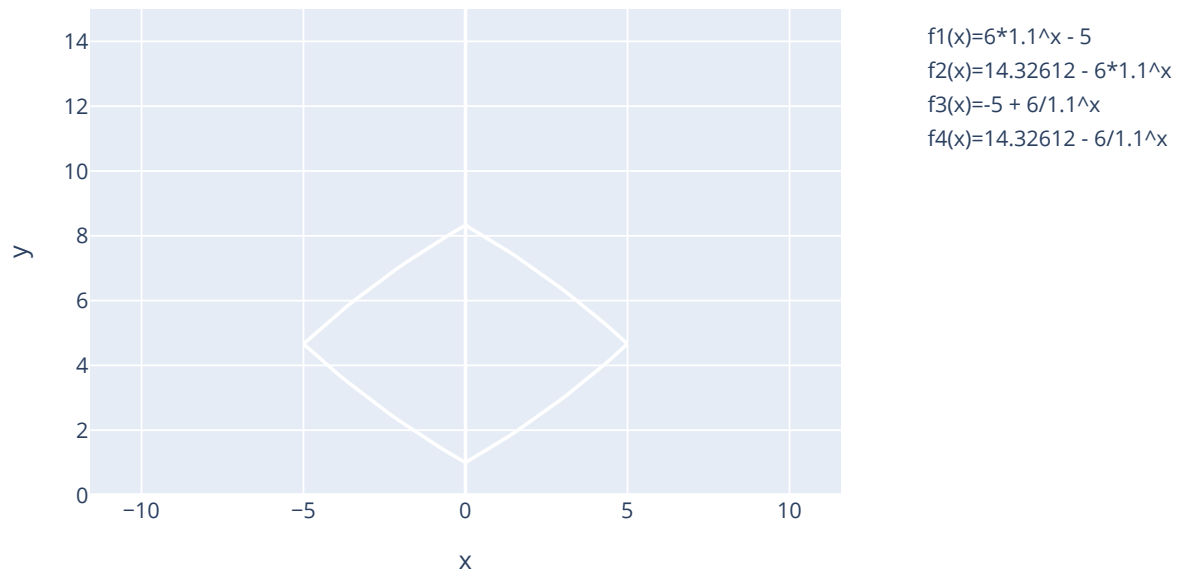
make_closed_fig(plot, a=a, b=1.1, h=0, k=-3, d=1, num_of_points=20, color= '#e06666')
plot.plot_funs(plot_title="Playing with mouth", img_name="frame3.png")

make_closed_fig(plot, a=a, b=1.1, h=1.5, k=-0.5, d=1, num_of_points=20, color= '#2986cc')
make_closed_fig(plot, a=a, b=1.1, h=-1.5, k=-0.5, d=1, num_of_points=20, color='#2986cc')

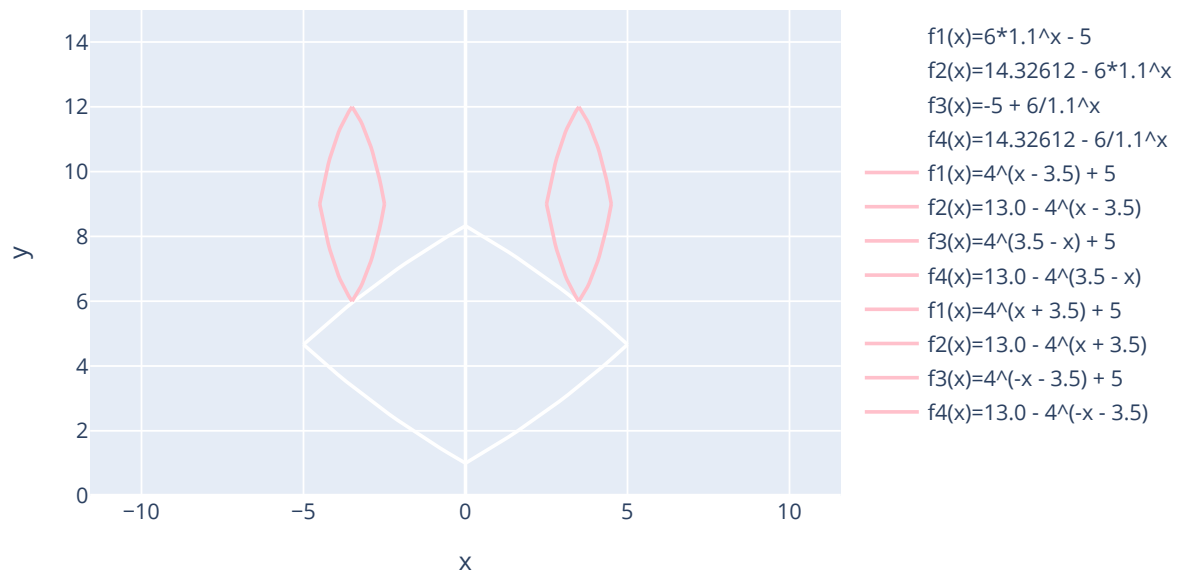
plot.plot_funs(plot_title="Playing with eyes", img_name="frame4.png")

make_closed_fig(plot, a=a, b=1.1, h=0, k=-1, d=0.25, num_of_points=20, color='pink')
plot.plot_funs(plot_title="Playing with nose", img_name="frame3.png")
```

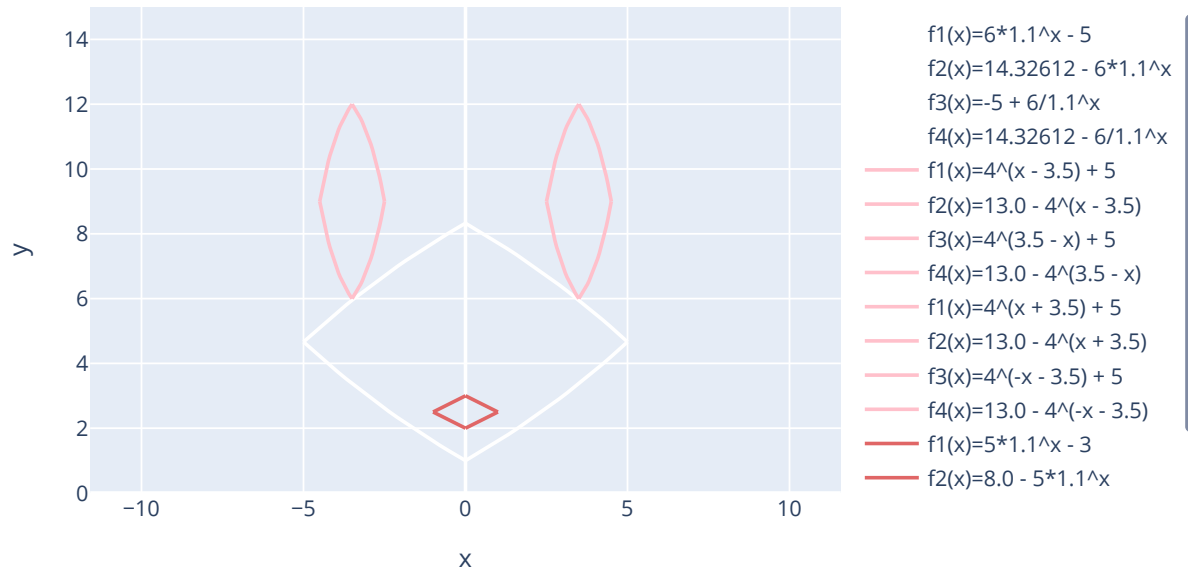
Playing with thehead



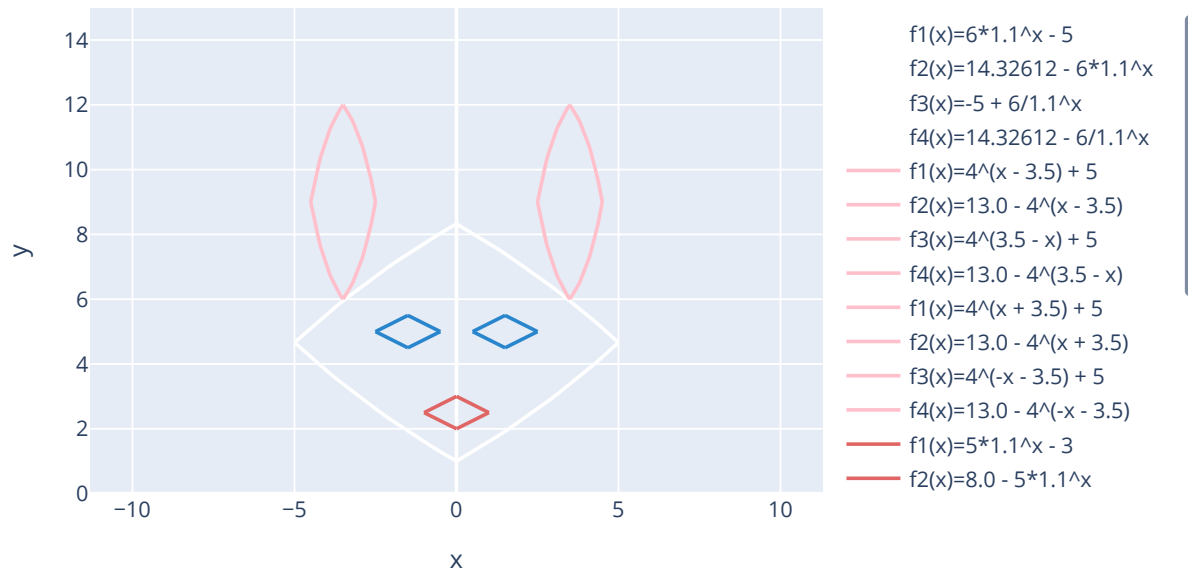
Playing with ears



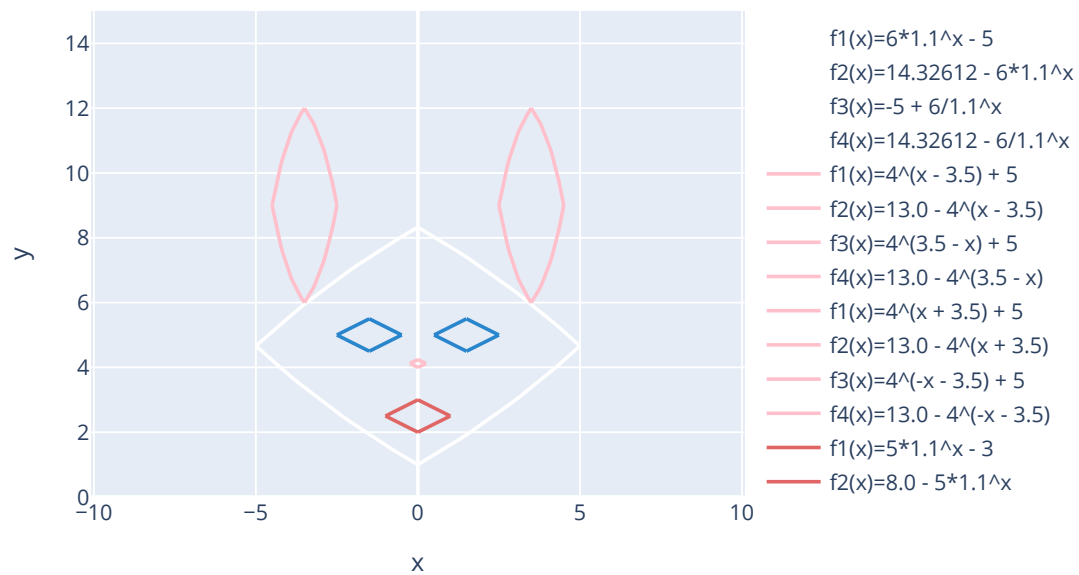
Playing with mouth



Playing with eyes



Playing with nose



```
In [ ]: !jupyter nbconvert --to html /content/drive/MyDrive/Colab_Notebooks/Untitled0.ipynb
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In [ ]:
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